Auction with Secret Reserve Price

Contents

1 Model Outline

- One seller, N bidders.
- Seller's cost is either c_L or c_H with $\Pr(c_L) = \psi$.
- Buyer's value is i.i.d. distributed according to some F supported on $[c_L, 1]$, with density f.
- **First stage:** an English auction with initially hidden reserve price, which is revealed conditional on being met.
- **Second stage:** if the English auction fails, one bidder is randomly drawn to make a TIOLI (take-it-or-leave-it) offer to the seller. An individual bidder is drawn with probability $\delta \leq 1/N$.

Notation. $\underline{r}$ and $\bar{r}$: endogenous lower and upper support of the low-cost seller's mixed strategy (chosen in equilibrium, not drawn); r_H: high-cost seller's reserve; $\rho \in \Delta[\underline{r}, \bar{r}]$: cdf of the reserve r actually played on the path; β: symmetric stage-1 bidding strategy; $\tilde{\psi}$: belief that the seller is low-cost.

2 Preliminary Analysis

2.1 Second-stage belief and offer

If the chosen buyer has a belief $\tilde{\psi}$, he offers price c_H if and only if

$$v - c_H \geq \tilde{\psi}(v - c_L) \quad \Longleftrightarrow \quad v \geq \frac{c_H - \tilde{\psi} c_L}{1 - \tilde{\psi}}. \tag{1}$$

2.2 Equilibrium structure

Consider an equilibrium where the high-cost seller plays a pure strategy r_H and the low-cost seller plays a mixed strategy denoted by $\rho \in \Delta[\underline{r}, \bar{r}]$.

Claim 1. *$r_H > \bar{r}$.*

For any increasing bidding function, the optimal reserve price for a high-cost seller is strictly above that of a low-cost seller.

2.3 High-cost seller

When the bid exceeds $\bar{r}$ but the reserve is not met, bidders believe the seller is high-cost and bid according to $\beta(v) = v - \delta(v - c_H)$. Once a bidder has $\beta(v) \geq r_H$, the reserve is announced as met and people bid their own values. The high-cost seller's profit $\Pi_H(r_H)$ is

$$\Pi_H(r_H) = \int_{\beta^{-1}(r_H)}^{1} \int_{c_L}^{v} \big(\max\{r_H, s\} - c_H\big)\, d\left(\frac{F(s)}{F(v)}\right)^{N-1} dF(v)^N. \tag{2}$$

Simplifying,

$$\begin{aligned}
\Pi_H(r_H) &= N\int_{\beta^{-1}(r_H)}^{1}\Big[(r_H-c_H)F(r_H)^{N-1}+\int_{r_H}^{v}(s-c_H)\,dF(s)^{N-1}\Big]\,dF(v)\\
&= N\Bigg[(r_H-c_H)F(r_H)^{N-1}\big(1-F(\beta^{-1}(r_H))\big)+\int_{r_H}^{1}(s-c_H)\,dF(s)^{N-1}\\
&\qquad -F(\beta^{-1}(r_H))\int_{r_H}^{\beta^{-1}(r_H)}(s-c_H)\,dF(s)^{N-1}-\int_{\beta^{-1}(r_H)}^{1}F(v)(v-c_H)\,dF(v)^{N-1}\Bigg].
\end{aligned} \tag{3}$$

The first-order condition is

$$r_H-c_H=(1-\delta)\frac{1-F\left(\dfrac{r_H-\delta c_H}{1-\delta}\right)}{f\left(\dfrac{r_H-\delta c_H}{1-\delta}\right)}-\frac{1}{F(r_H)^{N-1}}\int_{r_H}^{(r_H-\delta c_H)/(1-\delta)}(v-c_H)\,dF(v)^{N-1}. \tag{4}$$

2.4 Belief after a failed English auction

If the English auction fails with final bid b, the buyer's belief is

$$\tilde{\psi}=\frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}. \tag{5}$$

3 Bidding in the First-Stage English Auction

Suppose a bidder with value v decides his bid by comparing $v-b$ against

$$\delta\max\left\{v-c_H,\ \frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}(v-c_L),\ 0\right\}. \tag{6}$$

His bid thus satisfies

$$b=v-\delta\max\left\{v-c_H,\ \frac{\psi\big(1-\rho(b)\big)}{1-\psi\rho(b)}(v-c_L),\ 0\right\}. \tag{7}$$

The right-hand side is weakly decreasing in b and bounded between $[v-\delta\max\{v-c_H,\psi(v-c_L)\},\ v-\delta\max\{v-c_H,0\}]$. The bidding strategy β given ρ is thus

$$\beta(v)=\begin{cases} v-\delta\psi(v-c_L) & \text{if } v\le \dfrac{\underline{r}-\delta\psi c_L}{1-\delta\psi},\\ h^{-1}(v) & \text{if } v\in\left(\dfrac{\underline{r}-\delta\psi c_L}{1-\delta\psi},\ v^*\right),\\ v-\delta(v-c_H) & \text{if } v\ge v^*,\end{cases} \tag{8}$$

where

$$h(b)\equiv\frac{(1-\psi\rho(b))b-\delta\psi(1-\rho(b))c_L}{1-\delta\psi-(1-\delta)\psi\rho(b)}, \tag{9}$$

i.e. $h(b)$ solves $v - b = \delta\tilde{\psi}(b)(v - c_L)$ with $\tilde{\psi}(b) = \psi(1-\rho(b))/(1-\psi\rho(b))$ from (5), and $v^* \in \big[(\underline{r} - \delta\psi c_L)/(1-\delta\psi),\ (c_H - \psi c_L)/(1-\psi)\big]$ is the solution to

$$v - c_H = \frac{\psi\big(1 - \rho((1-\delta)v + \delta c_H)\big)}{1 - \psi\rho((1-\delta)v + \delta c_H)}(v - c_L) \quad\iff\quad \rho\big((1-\delta)v + \delta c_H\big) = \frac{\psi(v - c_L) - (v - c_H)}{\psi(c_H - c_L)}. \tag{10}$$

4 Low-Cost Seller Profit

It suffices to focus on the low-type seller's profit. Suppose it sets reserve price $\underline{r}$.

4.1 When the auction fails

The English auction *fails* when every bidder stops before the reserve $\underline{r}$ is met, so the sale moves to stage 2. With symmetric bidding β, that requires all N values to satisfy $\beta(v) \leq \underline{r}$, which occurs with probability

$$\Pr(\text{fail}) = F\left(\frac{\underline{r} - \delta\psi c_L}{1 - \delta\psi}\right)^N. \tag{11}$$

The seller may still earn revenue in stage 2. One bidder is drawn to make a take-it-or-leave-it offer with probability δ; across N bidders this contributes a factor δN. The selected buyer is unsure of the seller's cost and, by (1), offers c_H only if $v - c_H$ beats the expected gain from dealing with a low-cost seller. In the expression below we use the cutoff implied by prior belief ψ, $v \geq (c_H - \psi c_L)/(1-\psi)$. If the seller is in fact low-cost, serving the good still costs only c_L, so each accepted offer at c_H yields profit $c_H - c_L$.

Conditional on stage 1 failure, all values are relatively low—the maximum is at most $\beta^{-1}(\underline{r})$—yet a single bidder can still have a high enough v to offer c_H. Profit in this event is therefore

$$\Pi_{\text{fail}} = \delta N(c_H - c_L) \times \Pr\left(v_1 \geq \frac{c_H - \psi c_L}{1-\psi} \,\middle|\, v_{\max} \leq \beta^{-1}(\underline{r})\right), \tag{12}$$

Write $v_\psi \equiv (c_H - \psi c_L)/(1-\psi)$ and $z \equiv \beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1-\delta\psi)$. With i.i.d. values, (12) equals

$$\Pi_{\text{fail}} = \delta N(c_H - c_L)\ \Pr(\text{fail})\left(1 - \frac{F(v_\psi)}{F(z)}\right) = \delta N(c_H - c_L)\,\big(F(z) - F(v_\psi)\big)\,F(z)^{N-1}. \tag{13}$$

The second line uses (11) for $\Pr(\text{fail}) = F(z)^N$. Raising $\underline{r}$ makes stage 1 failure more likely because $F\big((\underline{r} - \delta\psi c_L)/(1-\delta\psi)\big)^N$ increases. The bracket in (13) also increases in $\underline{r}$: conditional on failure, a higher reserve means the surviving values can be larger, so it is more likely that some bidder still offers c_H in stage 2.

4.2 When the auction succeeds

The auction *succeeds* when at least one bidder is willing to push the bid above $\underline{r}$, so the reserve is met and the object is sold in stage 1. This requires $v_{\max} > \beta^{-1}(\underline{r})$, which occurs with probability $1 - F\big((\underline{r} - \delta\psi c_L)/(1-\delta\psi)\big)^N$. Once the reserve is met, bidders bid truthfully in the English auction. The bidder with the highest value v wins. The runner-up has the second-highest value s; the winner pays $\max\{\underline{r}, s\}$ (the reserve if $s < \underline{r}$, otherwise s). The seller's margin is therefore $\max\{\underline{r}, s\} - c_L$. On the low bidding branch, $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1-\delta\psi) > \underline{r}$ when $\underline{r} > c_L$, so every winning type satisfies $v > \underline{r}$ and the payment splits into two cases: the winner pays $\underline{r}$ when $s < \underline{r}$, and pays s when $s > \underline{r}$. Averaging over the winner's type v and the second-order statistic s gives

$$\Pi_{\text{succ}} = \int_{\beta^{-1}(\underline{r})}^{1} \int_{c_L}^{v} \big(\max\{\underline{r}, s\} - c_L\big)\, d\bigg(\frac{F(s)}{F(v)}\bigg)^{N-1} dF(v)^N, \tag{14}$$

where $d(F(s)/F(v))^{N-1}$ is the distribution of s conditional on $v_{\max} = v$, and $dF(v)^N = NF(v)^{N-1}f(v)\,dv$ is the density of the maximum. Integrating by parts yields the equivalent form

$$\Pi_{\text{succ}} = N\bigg\{(\underline{r} - c_L)F(\underline{r})^{N-1}\big(1 - F(\beta^{-1}(\underline{r}))\big) + \int_{\beta^{-1}(\underline{r})}^{1} \int_{\underline{r}}^{v} (s - c_L)\, dF(s)^{N-1}\, dF(v)\bigg\}. \tag{15}$$

The first term collects states where the runner-up's value is below $\underline{r}$, so the sale price equals the reserve. The double integral collects states where $s > \underline{r}$, so the winner pays the runner-up's value s (the usual second-price payment in an English auction).

4.3 Total profit and $\underline{r}^*$

Taken together, the low-type's profit is

$$\begin{aligned}\Pi(\underline{r}) \equiv{}& \delta N(c_H - c_L)\big(F(z) - F(v_\psi)\big)\, F(z)^{N-1} \\ &+ N\bigg\{(\underline{r} - c_L)F(\underline{r})^{N-1}\big(1 - F(\beta^{-1}(\underline{r}))\big) + \int_{\beta^{-1}(\underline{r})}^{1} \int_{\underline{r}}^{v} (s - c_L)\, dF(s)^{N-1}\, dF(v)\bigg\}.\end{aligned} \tag{16}$$

The formula $\Pi(r)$ attributes stage-2 payoffs using the prior ψ on every failure path. That matches announced reserves on the equilibrium path up to the lower support, but not when the seller raises the reserve above $\underline{r}$ and failure reveals bids above $\underline{r}$.

Claim 2. *The equilibrium lower support $\underline{r}^*$ is the argmax of $\Pi(\cdot)$ in (16).*

4.4 Proof of Claim 2

Proof. Let $\underline{r}^* \in \arg\max \Pi(\cdot)$. We show that any other lower support $\underline{r} \neq \underline{r}^*$ cannot be optimal.
Downward deviation. If $\underline{r} > \underline{r}^*$, stage-2 buyers keep the prior ψ, so (16) is exact and $\Pi(\underline{r}^*) > \Pi(\underline{r})$.
Upward deviation. If $\underline{r} < \underline{r}^*$, let $\tilde{\psi}(\beta(v)) \equiv \psi(1 - \rho(\beta(v)))/(1 - \psi\rho(\beta(v)))$ as in (5). The failure branch from announcing $\underline{r}^*$ is

$$\begin{aligned}&\delta N(c_H - c_L) \int_{c_L}^{\beta^{-1}(\underline{r})} 1 - \frac{F\big(\frac{c_H - \psi c_L}{1-\psi}\big)}{F(v)}\, dF(v)^N \\ &\quad + \delta N(c_H - c_L) \int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(\underline{r}^*)} 1 - \frac{F\big(\frac{c_H - \tilde{\psi}(\beta(v))c_L}{1-\tilde{\psi}(\beta(v))}\big)}{F(v)}\, dF(v)^N.\end{aligned} \tag{17}$$

The success branch is the same as in (15) with $\underline{r}^*$ in place of $\underline{r}$. For $v \in [\beta^{-1}(\underline{r}), \beta^{-1}(\underline{r}^*)]$, $\tilde{\psi}(\beta(v)) < \psi$, so $F\big(\frac{c_H - \tilde{\psi}(\beta(v))c_L}{1-\tilde{\psi}(\beta(v))}\big) < F\big(\frac{c_H - \psi c_L}{1-\psi}\big)$ and the failure branch above exceeds the failure term in $\Pi(\underline{r}^*)$. Hence announcing $\underline{r}^*$ yields profit at least $\Pi(\underline{r}^*) > \Pi(\underline{r})$.
Conclusion. Either deviation is profitable, so $\underline{r}^*$ is the optimal lower support. $\square$

The low-cost seller's equilibrium payoff is $\Pi(\underline{r}^*)$. On the low branch, $\beta^{-1}(\underline{r}) = (\underline{r} - \delta\psi c_L)/(1 - \delta\psi)$.

5 Low-Cost Seller: Optimal Support and Mixed Strategy

This section completes the low-cost seller's problem. Claim 2 pins the equilibrium lower support to the reserve that maximizes $\Pi(\cdot)$ in (16). We characterize that maximizer, extend payoffs to reserves above the lower support on the mixing path, and construct the reserve distribution from the resulting indifference condition.

5.1 Optimal lower support

The lower support $\underline{r}^*$ maximizes $\Pi(\underline{r})$ in (16). Imposing $\mathrm{d}\Pi(\underline{r})/\mathrm{d}\underline{r} = 0$ and rearranging yields the first-order condition

$$\begin{aligned}\underline{r}^* - c_L = (1-\delta\psi)\,\frac{1-F\big(\beta^{-1}(\underline{r}^*)\big)}{f\big(\beta^{-1}(\underline{r}^*)\big)} \\ + \frac{1}{F(\underline{r}^*)^{N-1}}\Bigg[\delta N(c_H-c_L)\,F\big(\beta^{-1}(\underline{r}^*)\big)^{N-2}\Big(NF\big(\beta^{-1}(\underline{r}^*)\big)-(N-1)F\Big(\frac{c_H-\psi c_L}{1-\psi}\Big)\Big) \\ - \int_{\underline{r}^*}^{\beta^{-1}(\underline{r}^*)}(s-c_L)\,dF(s)^{N-1}\Bigg].\end{aligned} \tag{18}$$

The derivation is in Appendix A.

5.2 Profit and indifference on the mixing path

In equilibrium the low-cost seller mixes a reserve $r \in [\underline{r}, \bar{r}]$ with cdf ρ, where $\underline{r}$ and $\bar{r}$ are the endogenous bounds of the support. If the realized reserve exceeds $\underline{r}$, stage 1 failure can occur with $v_{\max}$ between $\beta^{-1}(\underline{r})$ and $\beta^{-1}(r)$; bids then reveal $\rho(\beta(v)) > 0$ and beliefs update via (5). On the failure path, beliefs split at $\beta^{-1}(\underline{r})$: the prior ψ applies below that threshold and $\tilde{\psi}(\beta(v))$ applies above it until $\beta^{-1}(r)$. Payoff from announcing $r > \underline{r}$ is

$$\begin{aligned}\hat{\Pi}(r) = \delta N(c_H-c_L)\Big[\Big(F\big(\beta^{-1}(\underline{r})\big)-F\Big(\frac{c_H-\psi c_L}{1-\psi}\Big)\Big)F\big(\beta^{-1}(\underline{r})\big)^{N-1} \\ + F\big(\beta^{-1}(r)\big)^N - F\big(\beta^{-1}(\underline{r})\big)^N - N\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)} F\Big(\frac{(1-\psi\rho(\beta(v)))c_H-\psi(1-\rho(\beta(v)))c_L}{1-\psi}\Big) \\ \times F(v)^{N-2}\,dF(v)\Big] \\ + N\Big\{(r-c_L)F(r)^{N-1}\big(1-F(\beta^{-1}(r))\big) \\ + \int_{\beta^{-1}(r)}^{1}\int_{r}^{v}(s-c_L)\,dF(s)^{N-1}\,dF(v)\Big\}.\end{aligned} \tag{19}$$

Appendix B derives the middle integral. When $r = \underline{r}$, the middle terms vanish and (19) coincides with (16).

On the equilibrium path $\underline{r} = \underline{r}^*$, so the seller is indifferent across the support:

$$\hat{\Pi}(r) = \Pi(\underline{r}^*) \qquad \forall\, r \in [\underline{r}, \bar{r}]. \tag{20}$$

5.3 Construction of the mixing distribution

Differentiating (20) on the low branch yields an ordinary differential equation for $h(r) \equiv \beta^{-1}(r)$ and the mixing cdf $\rho(r)$; the algebra is in Appendix C. On the low branch, $h(r)$ coincides with (9) at $b = r$:

$$h(r) = \frac{(1-\psi\rho(r))r - \delta\psi(1-\rho(r))c_L}{1-\delta\psi-(1-\delta)\psi\rho(r)}, \tag{21}$$

with inverse

$$\rho(r) = \frac{-h(r) + r + \delta\psi(h(r) - c_L)}{\psi\big(-h(r) + r + \delta(h(r) - c_L)\big)}. \tag{22}$$

Indifference on the low branch is equivalent to

$$h'(r) = \frac{F(r)^{N-1}\big(1 - F(h(r))\big)}{f(h(r))} \times \mathcal{B}(r, h(r))^{-1}, \tag{23}$$

where $v_\rho(r, h)$ is the updated stage-2 cutoff when $\rho(r) > 0$,

$$v_\rho(r,h) \equiv \frac{-(h-r)c_L + \delta c_H(h - c_L)}{-(h-r) + \delta(h-c_L)},$$

and

$$\mathcal{B}(r,h) = (h - c_L)F(h)^{N-1} - \delta N(c_H - c_L)\big(F(h) - F(v_\rho(r,h))\big)F(h)^{N-2} - \int_r^h F(v)^{N-1}\,dv. \tag{24}$$

Claim 3. *Suppose F and f are continuous on $[c_L, 1]$ with $f > 0$. Let $\underline{r}^*$ satisfy (18) with $h^* = \beta^{-1}(\underline{r}^*)$ and $\rho(\underline{r}^*) = 0$ on the low branch. Then (23) admits a local solution $r \mapsto h(r)$ through $\big(\underline{r}^*, h^*\big)$.*

Proof. The right-hand side of (23) is continuous near $\big(\underline{r}^*, h^*\big)$ and $f(h^*) > 0$. By Peano's theorem, a local solution exists provided $\mathcal{B}(\underline{r}^*, h^*) > 0$. At $\rho(\underline{r}^*) = 0$ we have $v_\rho(\underline{r}^*, h^*) = (c_H - \psi c_L)/(1-\psi)$; Appendix C.3 shows that (18) implies $\mathcal{B}(\underline{r}^*, h^*) > 0$. ☐

A local solution is enough to define a mixed strategy on an interval $[\underline{r}^*, r_1]$: given h solving (23) on that interval, (22) pins down ρ, and (20) holds by construction on the low branch. For $r > \beta(v^*)$, $\beta^{-1}(r)$ is known in closed form and (38) in Appendix C extends ρ on the upper part of $[\underline{r}, \bar{r}]$. Together with $\bar{r} < r_H$ (Claim 1), this yields a mixed strategy equilibrium for the low-cost seller with payoff $\Pi(\underline{r}^*)$.

6 Case of Public Reserve Price

If the equilibrium is separating, the low-cost seller sets a reserve price given by

$$r_L - c_L = (1-\delta)\frac{1 - F\left(\dfrac{r_L - \delta c_L}{1-\delta}\right)}{f\left(\dfrac{r_L - \delta c_L}{1-\delta}\right)} - \frac{1}{F(r_L)^{N-1}}\int_{r_L}^{(r_L - \delta c_L)/(1-\delta)} (v - c_L)\,dF(v)^{N-1}. \tag{25}$$

whereas the high-cost seller sets reserve price weakly above r_H, solution to

$$r_H - c_H = (1-\delta)\frac{1 - F\left(\dfrac{r_H - \delta c_H}{1-\delta}\right)}{f\left(\dfrac{r_H - \delta c_H}{1-\delta}\right)} - \frac{1}{F(r_H)^{N-1}}\int_{r_H}^{(r_H - \delta c_H)/(1-\delta)} (v - c_H)\,dF(v)^{N-1}, \tag{26}$$

that is enough to deter low-type mimicking.

A FOC for $\underline{r}^*$

This appendix derives (18) from (16).

A.1 Derivative of the failure-region term

$$\begin{aligned}
&\frac{d}{d\underline{r}}\left[\delta N(c_H-c_L)\times\left(F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-1}\right]\\
&=\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-1}\\
&\quad+(N-1)\frac{1}{1-\delta\psi}\left(F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)\\
&=\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2}\left[NF\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-(N-1)F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right].
\end{aligned}\tag{27}$$

A.2 Simplifying the double integral

$$\begin{aligned}
&\int_{\beta^{-1}(\underline{r})}^{1}\int_{\underline{r}}^{v}(s-c_L)\,dF(s)^{N-1}\,dF(v)\\
&=\left[F(v)\int_{\underline{r}}^{v}(s-c_L)\,dF(s)^{N-1}\right]_{\beta^{-1}(\underline{r})}^{1}-\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}\\
&=\int_{\underline{r}}^{1}(s-c_L)\,dF(s)^{N-1}-F\big(\beta^{-1}(\underline{r})\big)\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}-\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}.
\end{aligned}\tag{28}$$

Derivative of the first term.

$$\frac{d}{d\underline{r}}\int_{\underline{r}}^{1}(s-c_L)\,dF(s)^{N-1}=-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r}).\tag{29}$$

Derivative of the second term.

$$\begin{aligned}
&\frac{d}{d\underline{r}}\left(F\big(\beta^{-1}(\underline{r})\big)\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\right)\\
&=f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\\
&\quad+(N-1)F\big(\beta^{-1}(\underline{r})\big)\Big[\big(\beta^{-1}(\underline{r})-c_L\big)F\big(\beta^{-1}(\underline{r})\big)^{N-2}f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}-(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\Big].
\end{aligned}\tag{30}$$

Derivative of the third term.

$$\frac{d}{d\underline{r}}\int_{\beta^{-1}(\underline{r})}^{1}F(v)(v-c_L)\,dF(v)^{N-1}=-\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}(N-1)\big(\beta^{-1}(\underline{r})-c_L\big)F\big(\beta^{-1}(\underline{r})\big)^{N-1}f\big(\beta^{-1}(\underline{r})\big).\tag{31}$$

The derivative of the double integral is therefore

$$-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big)-f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}. \quad (32)$$

Derivative of the bracket term.

$$\begin{aligned}&\frac{d}{d\underline{r}}\left[(\underline{r}-c_L)F(\underline{r})^{N-1}\big(1-F(\beta^{-1}(\underline{r}))\big)\right]\\ &=\begin{bmatrix}F(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big)\\ \quad+(N-1)(\underline{r}-c_L)\big(1-F(\beta^{-1}(\underline{r}))\big)f(\underline{r})\end{bmatrix}F(\underline{r})^{N-2}.\\ &\qquad-(\underline{r}-c_L)F(\underline{r})f\big(\beta^{-1}(\underline{r})\big)\frac{1}{1-\delta\psi}\end{aligned} \quad (33)$$

A.3 Full FOC and simplification

Taken together, $\mathrm{d}\Pi(\underline{r})/\mathrm{d}\underline{r}=0$ is

$$\begin{aligned}0&=\delta N(c_H-c_L)\frac{1}{1-\delta\psi}f\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)F\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)^{N-2}\\ &\quad\times\left[NF\left(\frac{\underline{r}-\delta\psi c_L}{1-\delta\psi}\right)-(N-1)F\left(\frac{c_H-\psi c_L}{1-\psi}\right)\right]\\ &\quad+N\begin{bmatrix}F(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big)\\ \quad+(N-1)(\underline{r}-c_L)\big(1-F(\beta^{-1}(\underline{r}))\big)f(\underline{r})\end{bmatrix}F(\underline{r})^{N-2}\\ &\qquad-(\underline{r}-c_L)F(\underline{r})f\big(\beta^{-1}(\underline{r})\big)\frac{1}{1-\delta\psi}\\ &\quad+N\Big[-(N-1)(\underline{r}-c_L)F(\underline{r})^{N-2}f(\underline{r})\big(1-F(\beta^{-1}(\underline{r}))\big)\\ &\qquad-f\big(\beta^{-1}(\underline{r})\big)\frac{d\beta^{-1}(\underline{r})}{d\underline{r}}\int_{\underline{r}}^{\beta^{-1}(\underline{r})}(s-c_L)\,dF(s)^{N-1}\Big].\end{aligned} \quad (34)$$

The $(N-1)(\underline{r}-c_L)$ terms from the bracket and double-integral derivatives cancel. Solving the remaining expression for $\underline{r}-c_L$ gives (18).

B Profit for $r>\underline{r}$

Write $v_\psi\equiv(c_H-\psi c_L)/(1-\psi)$ and $v_{\tilde{\psi}}(v)\equiv\big((1-\psi\rho(\beta(v)))c_H-\psi(1-\rho(\beta(v)))c_L\big)/(1-\psi)$. The stage-2 failure block in (19) is

$$\begin{aligned}&\delta N(c_H-c_L)\left(F\big(\beta^{-1}(\underline{r})\big)-F(v_\psi)\right)F\big(\beta^{-1}(\underline{r})\big)^{N-1}\\ &\quad+\delta N(c_H-c_L)\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}\left(1-\frac{F\big(v_{\tilde{\psi}}(v)\big)}{F(v)}\right)dF(v)^N,\end{aligned} \quad (35)$$

plus the stage-1 success terms in (19). The first line matches (13) at $r=\underline{r}$. Since $\int(1-F_{\tilde{\psi}}/F)\,dF^N=N\int(F-F_{\tilde{\psi}})F^{N-2}\,dF$,

$$\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}\left(1-\frac{F_{\tilde{\psi}}}{F}\right)dF^N=F\big(\beta^{-1}(r)\big)^N-F\big(\beta^{-1}(\underline{r})\big)^N-N\int_{\beta^{-1}(\underline{r})}^{\beta^{-1}(r)}F_{\tilde{\psi}}\,F^{N-2}\,dF,$$

which yields the middle terms in (19).

C Mixed-strategy indifference and the ODE

C.1 Incremental indifference

On $[\underline{r}, \bar{r}]$, equilibrium requires $\hat{\Pi}(r) = \Pi(\underline{r}^*)$. The ψ block in (19) does not depend on r; with $\underline{r} = \underline{r}^*$ on path,

$$
\begin{aligned}
&\delta N(c_H - c_L)\Bigg[F\big(\beta^{-1}(r)\big)^N - F\big(\beta^{-1}(\underline{r}^*)\big)^N - N\int_{\beta^{-1}(\underline{r}^*)}^{\beta^{-1}(r)} F\bigg(\frac{(1-\psi\rho(\beta(v)))c_H - \psi(1-\rho(\beta(v)))c_L}{1-\psi}\bigg) F(v)^{N-2}\, dF(v)\Bigg] \\
&\quad + N\bigg\{(r - c_L)F(r)^{N-1}\big(1 - F(\beta^{-1}(r))\big) + \int_{\beta^{-1}(r)}^{1}\int_r^v (s - c_L)\, dF(s)^{N-1}\, dF(v)\bigg\} \\
&= N\bigg\{(\underline{r}^* - c_L)F(\underline{r}^*)^{N-1}\big(1 - F(\beta^{-1}(\underline{r}^*))\big) + \int_{\beta^{-1}(\underline{r}^*)}^{1}\int_{\underline{r}^*}^v (s - c_L)\, dF(s)^{N-1}\, dF(v)\bigg\}. \qquad (36)
\end{aligned}
$$

C.2 Derivative and the ODE

On the low branch, $\beta^{-1}(r) = h(r)$. Differentiating (36) in r gives

$$
\begin{aligned}
0 &= \delta N(c_H - c_L)\Bigg[F\big(h(r)\big)^{N-1} f\big(h(r)\big)\, h'(r) - F\bigg(\frac{(1-\psi\rho(r))c_H - \psi(1-\rho(r))c_L}{1-\psi}\bigg) F\big(h(r)\big)^{N-2} f\big(h(r)\big)\, h'(r)\Bigg] \\
&\quad + F(r)^{N-1}\big(1 - F(h(r))\big) - (r - c_L)F(r)^{N-1} f\big(h(r)\big)\, h'(r) \\
&\quad - f\big(h(r)\big)\, h'(r)\int_r^{h(r)} (s - c_L)\, dF(s)^{N-1}. \qquad (37)
\end{aligned}
$$

Collecting terms with $h'(r)$ in (37) and solving gives (23)–(24).

C.3 Link between $\mathcal{B}$ and the FOC

Let $h = \beta^{-1}(r)$ and $v_\psi = (c_H - \psi c_L)/(1-\psi)$. Integration by parts gives $\int_r^h (s - c_L)\, dF(s)^{N-1} = (h - c_L)F(h)^{N-1} - (r - c_L)F(r)^{N-1} - \int_r^h F(v)^{N-1}\, dv$. Multiplying (18) by $F(r)^{N-1}$ and substituting this identity yields, whenever $\rho(r) = 0$ (so $v_\rho(r, h) = v_\psi$),

$$
\mathcal{B}(r, h) = (1 - \delta\psi)\frac{1 - F(h)}{f(h)} F(r)^{N-1} + \delta N(c_H - c_L)\Big[(N-1)F(h)^{N-1} - (N-2)F(v_\psi)\, F(h)^{N-2}\Big].
$$

The first term is positive when $F(h) < 1$. The second is positive when $(N-1)F(h) > (N-2)F(v_\psi)$ (e.g. when $F(h) > F(v_\psi)$ and $N \geq 2$). In particular, if $\underline{r}^*$ satisfies (18) with $h^* = \beta^{-1}(\underline{r}^*)$ and $\rho(\underline{r}^*) = 0$, then $\mathcal{B}(\underline{r}^*, h^*) > 0$.

C.4 High reserves

For $r > \beta(v^*)$, $\beta^{-1}(r) = (r - \delta c_H)/(1-\delta)$ and (37) becomes

$$\begin{aligned}
0 = \delta N(c_H - c_L) &\left(F\left(\frac{r-\delta c_H}{1-\delta}\right) - F\left(\frac{(1-\psi\rho(r))c_H - \psi(1-\rho(r))c_L}{1-\psi}\right)\right) F\left(\frac{r-\delta c_H}{1-\delta}\right)^{N-2} f\left(\frac{r-\delta c_H}{1-\delta}\right)\frac{1}{1-\delta} \\
&+ F(r)^{N-1}\left(1 - F\left(\frac{r-\delta c_H}{1-\delta}\right)\right) \\
&- \frac{1}{1-\delta} f\left(\frac{r-\delta c_H}{1-\delta}\right)\left[\left(\frac{r-\delta c_H}{1-\delta} - c_L\right) F\left(\frac{r-\delta c_H}{1-\delta}\right)^{N-1} - \int_r^{(r-\delta c_H)/(1-\delta)} F(v)^{N-1}\, dv\right],
\end{aligned} \tag{38}$$

which pins down $\rho(r)$ on the upper part of the support.